

Understanding Oracle APEX

The Platform, the Workspace, and What It Can Do

Courses	CSCI 6950 - Advanced Database Design and Administration CSCI 4850
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Format	Reading only. No SQL and no exercises in this document.
Estimated time	20 to 30 minutes
Graded	No. Background reading before the hands-on work.

Part 1 | About This Document

Overview

This document explains what Oracle APEX is before you write a single line of SQL in it. It covers the product, the way it is put together, the parts of the environment you will open, and the features that make it worth learning.

There is no code here and nothing to run. Read it once, and the hands-on documents will make far more sense. When a tool surprises you later, the explanation is usually somewhere in this file.

Everything described here applies to any APEX workspace, whether your instructor provides one or you request a free one from Oracle.

What You Will Learn

- What Oracle APEX is, and what problem it was built to solve.
- How the pieces fit together, from your browser down to the database.
- What a workspace is, and how it differs from a schema and a user account.
- What each main area of the environment does.
- Which features you will meet in this course, and which ones come later.
- The vocabulary you need to read Oracle documentation without getting lost.

Learning Outcomes

After you read this document, you will be able to:

1. Describe Oracle APEX in one or two sentences to someone who has never seen it.
2. Explain where APEX runs and why nothing needs installing on your machine.
3. Tell the difference between a workspace, a schema, and a user.
4. Name the main areas of the environment and say what each one is for.
5. List several APEX features and explain what each one gives a developer.
6. Use the correct term when you search Oracle documentation or ask for help.

DOCUMENT MAP

Part	Title	What it covers
1	About This Document	Purpose and outcomes
2	What Oracle APEX Is	The product and the problem it solves
3	How the Pieces Fit Together	Browser, web layer, engine, database
4	The Workspace	Workspace, schema, user, and roles
5	Inside the Environment	The main areas and what they do
6	Features Worth Knowing	What APEX gives you as a developer
7	Why This Matters	Value for the course and for your career
8	Vocabulary	Terms you will meet in the documentation
9	Where to Learn More	Official sources and community
10	What Comes Next	How this connects to the hands-on work

Part 2 | What Oracle APEX Is

Oracle Application Express, usually shortened to APEX, is a low-code development platform for building web applications on top of an Oracle Database. You design an application through a browser interface, and APEX generates the working web pages for you.

The word low-code means you declare what you want rather than writing every line yourself. You point APEX at a table, choose a report style, and it produces a searchable, sortable, mobile-friendly page. You still write SQL, and you still make design decisions, but you do not hand-build the plumbing that connects a browser to a database.

The problem it solves

Building a small business application the traditional way takes a long time. You need a front end, a back end, a database layer, session handling, login, security, and a way to deploy all of it. Most of that work has nothing to do with the actual problem you were trying to solve.

APEX removes most of that work. The database already stores the data, so APEX puts the application in the same place and generates the rest. A working application can exist in an afternoon instead of a month.

Key characteristics

Characteristic	What it means for you
Browser based	You develop and you run applications in a browser. Nothing installs on your machine.
Included with Oracle Database	APEX is a feature of the database, not a separate product you buy.
Database centred	Your data and your application definition live in the same database.
Declarative	You choose settings in a builder. APEX writes the page for you.
SQL friendly	Almost every component is driven by a SQL query you control.
Multi user by design	A workspace supports many developers working on the same applications.
Responsive by default	Generated pages already work on a phone, a tablet, and a desktop.

What APEX is not

- It is not a separate database. APEX runs inside Oracle Database and depends on it.
- It is not a replacement for learning SQL. The better your SQL, the better your APEX applications.
- It is not limited to small tools. Organizations run production systems on it with thousands of users.
- It is not a general purpose website builder. It is aimed at data driven applications, not marketing sites.

One sentence version: Oracle APEX is a browser based, low-code platform that turns an Oracle Database into a working web application without a separate application stack.

Part 3 | How the Pieces Fit Together

APEX has fewer moving parts than most web platforms. Understanding the four layers below explains most of its behaviour, including why it feels fast and why everything is tied to the database.

Layer	What it does
Your browser	Sends requests and displays pages. No plugin and no local install are needed.
Web listener	Receives the request from the browser and passes it to the database. Oracle REST Data Services, known as ORDS, normally fills this role.
APEX engine	Code that lives inside the database. It reads the definition of your application and builds the page that gets sent back.
Oracle Database	Stores two things: the data your application works with, and the definition of the application itself.

The last point is the one that surprises people. When you build a page in APEX, you are not creating files on a server. You are creating rows in database tables. The APEX engine reads those rows every time somebody opens the page, and renders the result.

Why this matters: Because the application is stored as data, a database backup captures your application as well. Moving an application to another environment means exporting and importing it, not copying a folder of files.

What this design gives you

- The data never travels far. Processing happens next to where the data is stored.
- There is no separate application server to patch, configure, and keep in sync.
- Database security features apply to your application, because the application sits inside the database.
- Several developers can work in the same environment without setting up local machines.

Part 4 | The Workspace

A workspace is the container you log in to. It groups a set of users, one or more database schemas, and the applications built against them. Everything you do in this course happens inside one workspace.

The sign-in page asks for a workspace name, a username, and a password. Three fields instead of two is the first sign that a workspace is a real boundary and not just a label.

Workspace, schema, and user

These three terms get mixed up constantly. They are not the same thing.

Term	What it is	Think of it as
Workspace	A named area that holds users, schemas, and applications	The room you work in

Term	What it is	Think of it as
Schema	The database account that owns tables, views, and other objects	The filing cabinet in that room
User	An account that signs in to the workspace with a role attached	The person holding the key
Application	A set of pages that work together as one web application	A finished product built in that room

The most common confusion: A workspace can be linked to more than one schema. That is why the SQL Workshop has a Schema selector. If you create a table while the wrong schema is selected, the table is real, but it is not where you expect to find it.

Roles inside a workspace

Role	What that person can do
Workspace administrator	Manages users, monitors activity, and controls workspace settings
Developer	Builds applications and works with database objects. This is your role in the course.
End user	Runs the finished applications but cannot open the development tools

Where workspaces come from

- An instructor or an administrator creates one and gives you the credentials. This is the usual arrangement for a class.
- Oracle offers free workspaces for learning and evaluation, requested through the APEX website.
- An organization runs APEX on its own Oracle Database, or on an Oracle Cloud autonomous database, and creates workspaces internally.

The environment looks the same in all three cases. What changes is who administers it and how much storage you get.

Part 5 | Inside the Environment

After you sign in, a navigation bar runs across the top of every page. Each entry opens a different area of the environment.

Area	What it is for	When you use it
App Builder	Creating and editing applications and their pages	Later in the course, once the data exists
SQL Workshop	Working directly with the database: tables, data, and SQL	Constantly, and from your very first day
Team Development	Tracking features, bugs, and to-do items for a project	Group projects, if your instructor uses it
Gallery	Sample and starter applications you can install and study	When you want to see how something was built
Administration	Users, storage, activity reports, and workspace settings	Mostly a workspace administrator task

Inside the SQL Workshop

The SQL Workshop is where a database course lives. It holds five tools.

Tool	What it does	Reach for it when
Object Browser	A point-and-click view of tables, views, indexes, and other objects	You want to inspect structure or data without writing SQL
SQL Commands	Runs a single SQL statement and shows the result underneath	You are testing one query or one change
SQL Scripts	Stores many statements under a name and runs them together	You want to repeat a job or share it with someone
Utilities	Data loading, data generation, schema comparison, and reports	You need to move data in or out, or study a schema
RESTful Services	Publishes database data as web services other systems can call	Advanced work, and usually later in a program

The three you will use most: Object Browser to look, SQL Commands to test, SQL Scripts to repeat. The hands-on document covers all three in order.

Part 6 | Features Worth Knowing

You will not touch all of these this semester. Knowing the names is still useful, because they explain what the platform is capable of and they appear throughout Oracle documentation.

Working with data

Feature	What it gives you
Interactive Report	A results grid the end user can filter, sort, group, highlight, and download without help from a developer
Interactive Grid	The same idea, with editing. Users change rows directly in the grid.
Faceted search	Filter panels beside a report, similar to the filters on a shopping site
Charts	Bar, line, pie, and many other chart types driven by a SQL query
Data loading	Load a spreadsheet or a delimited file into a table through a guided wizard
Data generator	Create realistic sample data for a table so you can test with volume

Building the application

Feature	What it gives you
Page Designer	The main development screen, where pages, regions, and items are assembled
Create Application wizard	Generates a working multi-page application from your tables in a few clicks
Universal Theme	The default look and layout, already responsive and accessible
Forms and reports	Standard page types generated directly from a table or a query
Validations and processes	Rules that check user input, and logic that runs when a page is submitted
Quick SQL	Turns a short indented outline into full table creation statements

Security and integration

Feature	What it gives you
Authentication schemes	Different ways to prove who a user is, including database accounts and single sign-on
Authorization schemes	Rules that decide which users can see or do what, applied per page or per component
Session state	The way APEX remembers values as a user moves between pages
REST data sources	Use data from an external web service as though it were a local table
Email and file storage	Send mail and store uploaded documents from inside an application

Working as a team

- Applications export as a single file, which can be stored in version control alongside other project code.
- Working copies let two developers change the same application separately and merge the results.
- Team Development tracks features and bugs inside the workspace itself.
- Activity reports show who ran what and how long each page took.

Part 7 | Why This Matters

For this course

APEX gives you a real Oracle Database in a browser, with no setup and no local installation. Every SQL statement you write runs against a genuine Oracle instance, not a simulator. That makes it a fair place to learn design, querying, and administration.

It also shortens the gap between a schema and something you can see. A table becomes a working page in minutes, which makes the consequences of a design decision obvious much sooner.

For your career

- Oracle Database is widely used in finance, healthcare, government, education, and manufacturing. Familiarity with its tools is a practical advantage.
- APEX rewards strong SQL. The skill you build here transfers to any database role, whether or not you use APEX again.
- Low-code platforms are now common across the industry. Learning one well makes the next one easier.
- The habits this environment teaches, such as saving repeatable scripts and thinking about who owns a schema, apply everywhere.

A useful way to think about it: APEX is a fast way to see what a database can do. The database is the part worth mastering, and APEX keeps that part in the foreground instead of hiding it.

Part 8 | Vocabulary

These terms appear throughout Oracle documentation and in the rest of this course.

Term	Meaning
APEX	Oracle Application Express, the low-code platform described in this document
Workspace	The container that groups users, schemas, and applications
Schema	The database account that owns tables and other objects

Term	Meaning
ORDS	Oracle REST Data Services, the web layer that connects a browser to the database
Application	A set of pages that work together as one web application
Page	A single screen in an application
Region	A block of content on a page, such as a report, a chart, or a form
Item	A single field on a page, such as a text box or a date picker
Process	Logic that runs at a defined point, usually when a page is submitted
Session state	The values APEX remembers for a user as they move between pages
DDL	Data Definition Language, the commands that create and change structure
DML	Data Manipulation Language, the commands that work with the data itself
Commit	The command that makes your data changes permanent
Script	A saved file of SQL statements that runs as one job

Part 9 | Where to Learn More

Source	What you find there
apex.oracle.com	The product site, with tutorials, sample applications, and free workspace requests
Oracle APEX documentation	The official reference, including an application builder guide and a full SQL reference
Oracle LiveLabs	Free guided workshops that walk through building specific features
apex.world	The community site, with plugins, articles, and discussion
Oracle APEX YouTube channel	Short feature demonstrations and release overviews

Search results for APEX often mix several versions of the product. Check the version shown in the bottom corner of your own environment, and prefer documentation that matches it.

Part 10 | What Comes Next

The hands-on document takes over from here. It stays inside the SQL Workshop and walks through the three tools you just read about.

1. Inspect a schema with the Object Browser.
2. Create your first table and choose a data type for each column.

3. Add records, then query them with SELECT.
4. Filter, sort, and summarize the results.
5. Change records and change the table itself.
6. Turn the whole job into a saved SQL script that runs in one click.

Nothing in this document needs memorizing. Come back to it when a term in the environment does not make sense.

QUESTIONS

Contact your instructor first. For help with the environment, you can also contact the teaching assistant. Both addresses are on page 1.